



Founded 1982

The Hills Grammar School

2013
Higher School Certificate
Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen. Black pen is preferred.
- Board approved calculators may be used.
- A standard table of integrals is provided at the back of this paper.
- In Questions 11-14, show relevant mathematical reasoning and/or calculations.

Total marks – 70

Section I

10 marks

- Attempt Questions 1-10
- Answer on the Multiple Choice Answer Sheet provided.
- Allow about 15 minutes for this section.

Section II

60 marks

- Attempt Questions 11-14
- Answer in the writing booklets provided.
- Start a new booklet for each question.
- Allow about 1 hour and 45 minutes for this Section.

Section 1

10 marks

Attempt Questions 1-10

Allow approximately 20 min for this section

1. The graph of the function $f(x) = -x(x^2 - 1)(x^2 + 1)$ cuts the x -axis
- A. five times
- B. twice
- C. three times
- D. four times.
2. The solutions to the equation $2 \sin \frac{x}{2} = 1$ over the domain $0 \leq x \leq 2\pi$ are
- A. $\frac{\pi}{12}$ and $\frac{5\pi}{12}$
- B. $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$
- C. $\frac{\pi}{3}$ and $\frac{5\pi}{3}$
- D. $\frac{8\pi}{3}$ and $\frac{10\pi}{3}$.
3. If $y = e^{4x} \cos x$ then $\frac{dy}{dx}$ equals
- A. $-4e^{4x} \sin x$
- B. $4e^{4x} \cos x + e^{4x} \sin x$
- C. $4e^{4x} \sin x$
- D. $e^{4x} (4 \cos x - \sin x)$.
4. If $\int_m^1 (1-x) dx = 8$ and $m \geq 0$, then m is equal to
- A. 7
- B. 5
- C. 0
- D. -3.

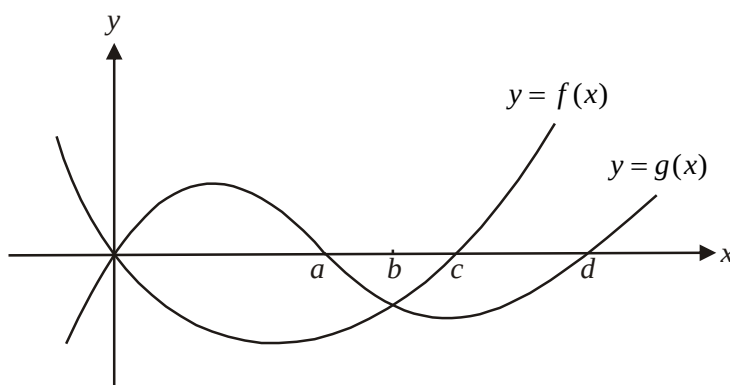
5. The displacement x , in metres, of a particle moving in a straight line is given by $x = \tan 3t$, $0 \leq t \leq \frac{\pi}{6}$ where t represents time in seconds. The acceleration, a , of the particle is given by

- A. $a = 9 \tan 3t$ C. $a = \frac{-2}{\cos^3 3t}$
 B. $a = 3 \sec^2 t$ D. $a = 18 \tan 3t \cdot \sec^2 3t$.

6. A ball is dropped from a height. The acceleration $a \text{ ms}^{-2}$ and the velocity $v \text{ ms}^{-1}$ of the ball t seconds after it is dropped is given by $a = 10 - \frac{v}{20}$. An expression for t in terms of v could be obtained by simplifying:

- A. $t = 20 \int \frac{1}{200 - v} dv$ C. $t = \int (\frac{v}{20} - 10) dv$
 B. $t = \int (10 - \frac{v}{20}) dv$ D. $t = \int (\frac{1}{10} - \frac{20}{v}) dv$.

7. The graphs of $y = f(x)$ and $y = g(x)$ are shown below.



The area enclosed by these two graphs is given by:

- A. $\int_0^b (g(x) - f(x)) dx$ C. $\int_0^a (g(x)) dx - \int_0^b f(x) dx$
 B. $\int_0^b (g(x) + f(x)) dx$ D. $\int_0^a (g(x)) dx - \int_0^c f(x) dx$.

8. Rhea made an error proving that $3^{2n} - 1$ is divisible by 8 (where n is an integer greater than 0) using mathematical induction. Part of the proof is shown below.

Step 1: test for $n = 1$, $3^{2(1)} - 1 = 8$, therefore divisible by 8.

Step 2: Assume the result true for $n = k$

$3^{2k} - 1 = 8P$ where P is an integer. Line 1

Hence $3^{2k} = 8P + 1$

To prove the result is true for $n = k + 1$

$3^{2(k+1)} - 1 = 8Q$ where Q is an integer. Line 2

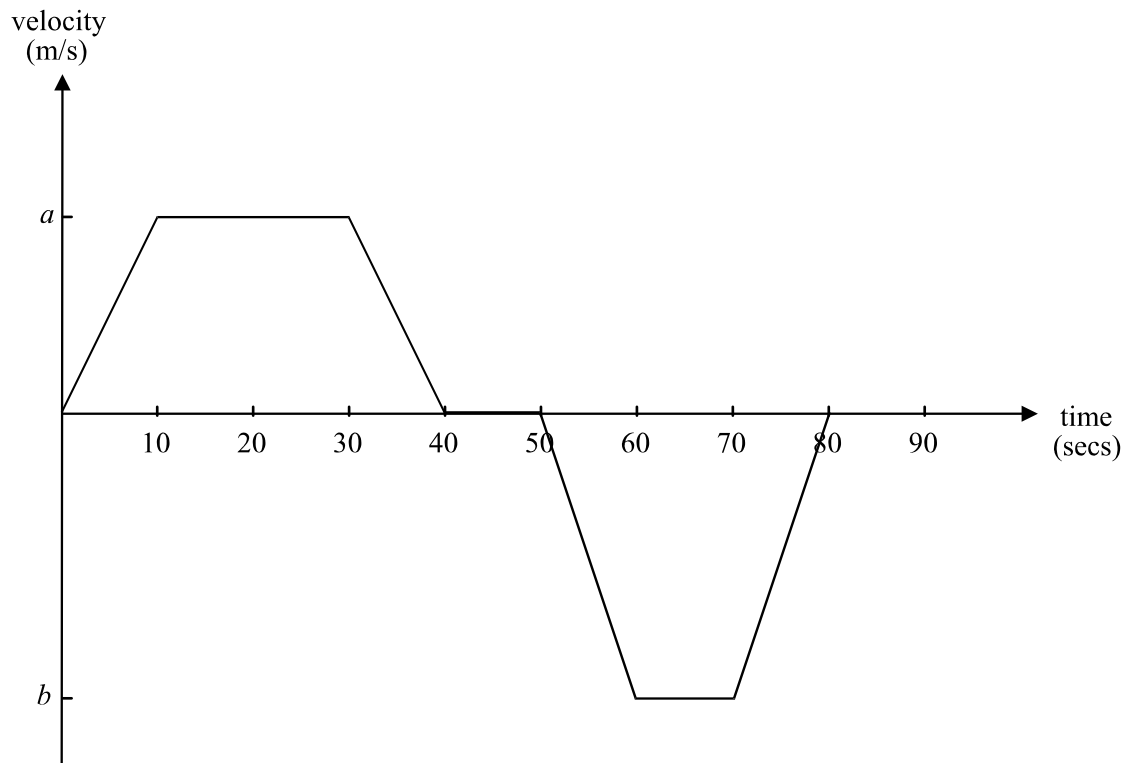
$$\begin{aligned}
 \text{LHS} &= 3^{2(k+1)} - 1 \\
 &= 3^{2k} \times 3^2 - 1 \\
 &= (8P + 1) \times 3^2 - 1 && \text{Line 3} \\
 &= 72P + 1 - 1 && \text{Line 4} \\
 &= 72P \\
 &= 8(9P) \\
 &= 8Q \\
 &= \text{RHS}
 \end{aligned}$$

Which line did Rhea make an error?

- | | |
|---|---|
| <p>A. Line 1</p> <p>B. Line 3</p> | <p>C. Line 2</p> <p>D. Line 4</p> |
|---|---|
9. A particle moving in simple harmonic motion starts from rest at distance 6 metres from the centre of oscillation. The period is 4π seconds. What is the minimum time taken to move to a point 3 metres from the origin?

- | | |
|---|---|
| <p>A. $t = \frac{\pi}{3}$</p> <p>B. $t = \frac{\pi}{6}$</p> | <p>C. $t = \frac{2\pi}{3}$</p> <p>D. $t = \frac{5\pi}{6}$</p> |
|---|---|

10. A man makes a return trip along the same route by car from his home to a letter box. The velocity-time graph describing his trip is shown below.



The initial acceleration of the car is 2 m/s^2 .

The values of a and b respectively are

- | | |
|-----------------|-------------------|
| A. 10 and -20 | C. 20 and -20 |
| B. 10 and -25 | D. 20 and -30 . |

End of Section 1

Section 11**60 marks****Attempt Questions 11-14****Allow approximately 1 hour 40 minutes for this section**Answer each question in a **separate writing booklet**.

Show all necessary working.

Diagrams are not drawn to scale.

Question 11 (15 marks)**Marks**

- | | | |
|-----|---|---|
| (a) | Find $\int \frac{e^x}{1+e^x} dx$. | 1 |
| (b) | State the domain and range of $y = 3\sin^{-1}\left(\frac{x}{2}\right)$. | 2 |
| (c) | The remainder when $P(x)$ is divided by $x^2 - 1$ is $3x - 4$. What is the remainder when $P(x)$ is divided by $x + 1$? | 2 |
| (d) | Differentiate with respect to x : $y = \ln(x^3 \sqrt{x^2 + 1})$. | 2 |
| (e) | Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{\tan 3x}$ showing appropriate working. | 2 |
| (f) | Solve the inequality $x \geq \frac{6}{x-1}$. | 3 |
| (g) | Write $\sqrt{3}\cos x - \sin x$ in the form $R\cos(x + \alpha)$ and hence find the general solution of the equation $\sqrt{3}\cos x - \sin x = 1$ | 3 |

Question 12 (15 marks)**Marks**

(a) $P(x) = 2x^3 - 4x^2 + 3x - 10$ has zeros α , β and γ . Determine the value of:

(i) $\alpha^2 + \beta^2 + \gamma^2$ 2

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2

(b) The function defined by $f(x) = x^3 - \ln(x+1)$ has one real root.

(i) Show that this root lies between $x = 0.8$ and $x = 0.9$. 1

(ii) Use the halving the interval method to approximate this root correct to two decimal places. 3

(c) Evaluate $\int_1^2 \sqrt{4-x^2} dx$ using the substitution $x = 2 \sin \theta$. 4

(d) Find the area between the curve $y = 2 \tan^{-1}\left(\frac{x}{2}\right)$ and the lines $x = 0$, $x = 2$ and the x axis. 3

Question 13 (15 marks)**Marks**

- (a) Consider the curve given by $y = \frac{x^2 + 9}{x}$.
- (i) Find any stationary points on the curve and determine their nature. 2
 - (ii) Find the vertical and oblique asymptotes of the curve. 2
 - (iii) Sketch the curve showing the above features. 2
- (b) Find the greatest coefficient in the expansion of $(7 + 3x)^{18}$. Leave your answer in factor form. 3
- (c) The coefficient of x^k in $(1 + x)^n$, where n is a positive integer, is denoted by c_k (where $c_k = {}^nC_k$).
- Show that
- $$c_0 + 2c_1 + 3c_2 + \dots + (n+1)c_n = (n+2)2^{n-1}.$$
- (Hint you will need to differentiate) 3
- (d) Prove using mathematical induction that $n^3 - n$ is divisible by 6 for all integers $n \geq 1$. 3

Question 14 (15 marks)**Marks**

(a)

(i) Evaluate $\int_{-2\sqrt{3}}^2 \frac{1}{4+x^2} dx$. 2

(ii) Prove: $\tan^{-1} a - \tan^{-1} b = \tan^{-1} \left(\frac{a-b}{1+ab} \right)$. 2

(b) A particle P is moving in simple harmonic motion about the origin. The particle is initially at rest a units to the right of the origin. If v is the velocity, x is the displacement and $\frac{2\pi}{n}$ is the period of the motion of the particle P at time t :

(i) Show that $v^2 = n^2(a^2 - x^2)$. 2

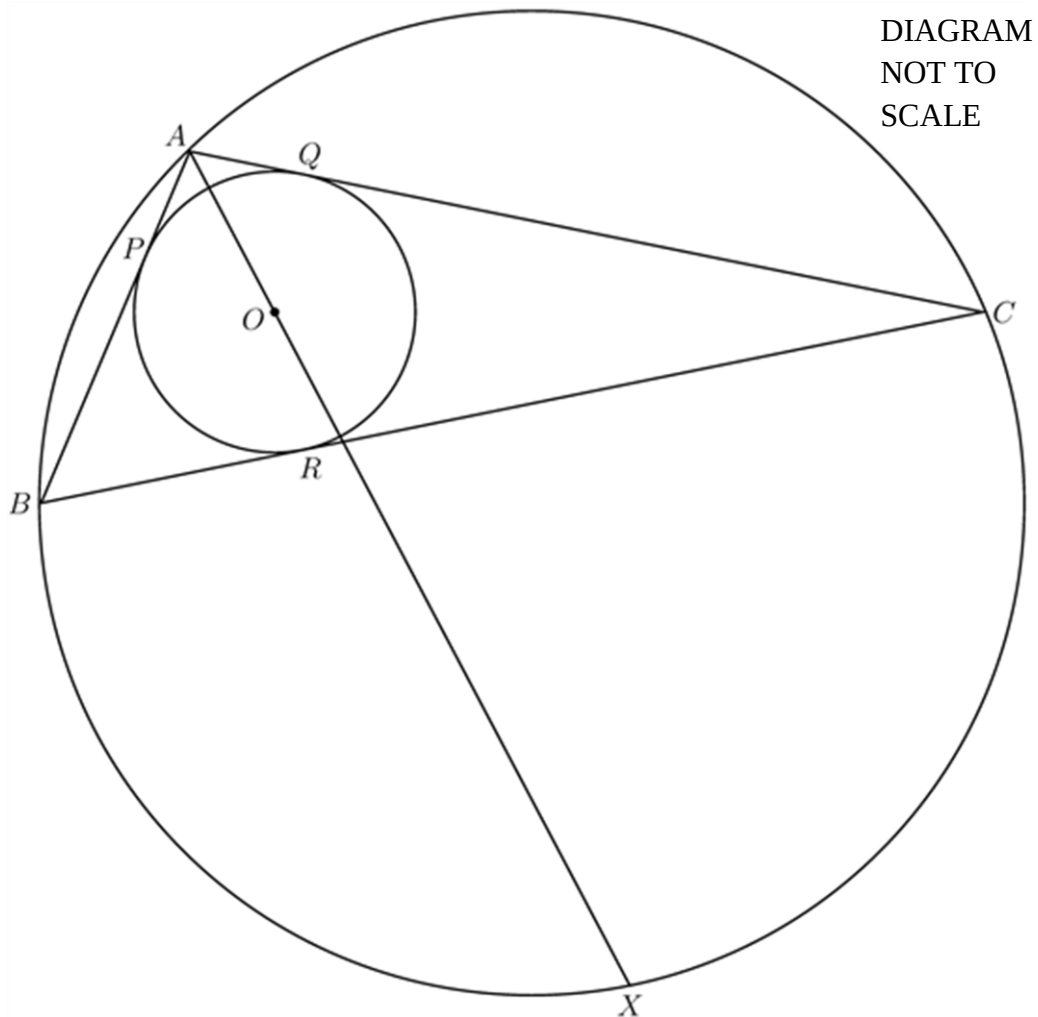
(ii) Hence show that $x = a \cos nt$ 2

(iii) Hence find an expression for v in terms of t . 1

(iv) Find the greatest speed of the particle P . 1

(v) Find the greatest magnitude of the acceleration of the particle P . 1

- (c) The circle inscribed in triangle ABC has centre O and meets the sides of the triangle ABC at P , Q and R . The circumcircle (ie. the circle that passes through each of the vertices of the triangle ABC) meets AO produced at X .



Let $\angle OAP = \alpha$ and $\angle OBR = \beta$.

Prove that:

- | | | |
|------|--------------------------------------|---|
| (i) | $\triangle OPA \equiv \triangle OQA$ | 2 |
| (ii) | X is the midpoint of arc BC . | 2 |

END OF ASSESMENT

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

ANSWER SHEET FOR MULTIPLE CHOICE SECTION

Student Exam number: _____

Teacher: _____

1. A ○ B ○ C ○ D ○

2. A ○ B ○ C ○ D ○

3. A ○ B ○ C ○ D ○

4. A ○ B ○ C ○ D ○

5. A ○ B ○ C ○ D ○

6. A ○ B ○ C ○ D ○

7. A ○ B ○ C ○ D ○

8. A ○ B ○ C ○ D ○

9. A ○ B ○ C ○ D ○

10. A ○ B ○ C ○ D ○

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MC Q

1, C
2, C
3, D
4, B
5, D

6, A
7, A
8, D
9, C
10, D

Q11.

$$(a) \int \frac{e^x}{1+e^{2x}} dx = \ln(1+e^x) + C \quad (1)$$

$$(b) y = 3 \sin^{-1}\left(\frac{x}{2}\right)$$

$$\text{Domain } -1 \leq \frac{x}{2} \leq 1$$

$$\therefore -2 \leq x \leq 2 \quad (1)$$

$$\text{Range } -\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2} \quad (1)$$

(c) From info:

$$P(x) = (x^2-1)Q(x) + 3x-4 \quad (1)$$

$$\text{Rem} = P(-1) = -7 \quad (1)$$

$$(d) y = \ln(x^3 \sqrt{x^2+1})$$

$$= 3\ln x + \ln(x^2+1)^{1/2} \quad (1)$$

$$y' = \frac{3}{x} + \frac{1}{2} \cdot \frac{2x}{x^2+1} \quad (1)$$

$$= \frac{3(x^2+1) + x^2}{x(x^2+1)}$$

$$= \frac{4x^2+3}{x(x^2+1)} \quad (1)$$

$$(e) \lim_{x \rightarrow 0} \frac{\sin 2x}{\tan 3x} = \frac{2}{3} \cdot \lim_{x \rightarrow 0} \frac{3x}{\tan 3x} \cdot \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$$

$$= \frac{2}{3} \cdot 1 \cdot 1$$

$$= \frac{2}{3} \quad (1)$$

$$(f) x \geq \frac{6}{x-1}, x \neq 1$$

$$x(x-1)^2 \geq 6(x-1) \quad (1)$$

$$x(x-1)^2 - 6(x-1) \geq 0$$

$$(x-1)(x^2-x-6) \geq 0$$

$$(x-1)(x-3)(x+2) \geq 0 \quad (1)$$

$$x = 1, 3, -2$$



$$\therefore \text{SS} = \{x \leq -2\} \cup \{1 \leq x \leq 3\} \cup \{x \geq 3\} \quad (1)$$

↑ NB

Q11 cont.

$$(g) \sqrt{3} \cos x - \sin x = 1$$

$$R = \sqrt{3+1} = 2$$

$$2 \cos(x+\alpha) = 1$$

$$\cos \alpha = \frac{\sqrt{3}}{2}, \sin \alpha = \frac{1}{2}$$

$$\therefore \tan \alpha = \frac{1}{\sqrt{3}} \quad \alpha = \frac{\pi}{6} \quad (1)$$

$$\therefore \cos\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$$

$$x + \frac{\pi}{6} = 2\pi n \pm \frac{\pi}{3} \quad (1)$$

$$x = 2\pi n + \pi/6 \text{ or}$$

$$2\pi n - \frac{\pi}{2}, \quad n \in \mathbb{Z} \quad (1)$$

Q12

(a) i) $P(x) = 2x^3 - 4x^2 + 3x - 10$

$$x^2 + \beta^2 + \gamma^2 = (x + \beta + \gamma)^2 - 2(x\beta + \beta\gamma + \gamma x)$$

$$= \left(-\frac{4}{2}\right)^2 - 2\left(\frac{3}{2}\right)$$

$$= 4 - 3$$

$$= 1$$

(ii) $\frac{1}{x} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + x\gamma + x\beta}{x\beta\gamma}$

$$= \frac{3/2}{-10/2}$$

$$= 3/10$$

(b) (i) $f(x) = x^3 - \ln(x+1)$

$$f(0.8) = -0.075 < 0$$

$$f(0.9) = 0.089 > 0$$

Since signs opp, root lies bet 0.8 and 0.9.

(ii) $x_1 = \frac{0.8+0.9}{2} = 0.85$

$$f(0.85) = -0.00106 < 0$$

Root bet 0.85 and 0.9

$$x_2 = \frac{0.85+0.9}{2} = 0.875$$

$$f(0.875) = 0.0413 > 0$$

∴ root lies bet 0.875 and 0.85

$$x_3 = \frac{0.875+0.85}{2} = 0.8625$$

$$f(0.8625) = 0.00196 > 0$$

∴ root bet 0.85 and 0.8625.

$$x_4 = \frac{0.85+0.8625}{2} = 0.85625$$

$$f(0.85625) = 0.000921 > 0$$

∴ root lies bet 0.85 and 0.85625

$$x_5 = \frac{0.85+0.85625}{2} = 0.853125$$

∴ $x \approx 0.85$ (2 d.p.)

(c) $I = \int_0^2 \sqrt{4-x^2} dx$

$$x = 2 \sin \theta$$

$$\frac{dx}{d\theta} = 2 \cos \theta$$

$$x = 1, \theta = \pi/6$$

$$x = 2, \theta = \pi/2$$

details

$$\therefore I = \int_{\pi/6}^{\pi/2} \sqrt{4-4\sin^2 \theta} \cdot 2 \cos \theta d\theta$$

$$= \int_{\pi/6}^{\pi/2} 2 \cos \theta \cdot 2 \cos \theta d\theta$$

$$= 4 \int_{\pi/6}^{\pi/2} \cos^2 \theta d\theta$$

$$= 2 \int_{\pi/6}^{\pi/2} 1 + \cos 2\theta d\theta = 2 \left[\theta + \frac{1}{2} \sin 2\theta \right]_{\pi/6}^{\pi/2}$$

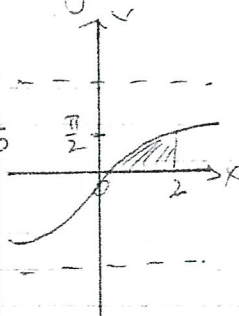
$$= 2 \left[\left(\frac{\pi}{2} + \frac{1}{2} \sin \pi \right) - \left(\frac{\pi}{6} + \frac{1}{2} \sin \frac{\pi}{3} \right) \right]$$

$$= 2 \left[\frac{\pi}{2} - \frac{\pi}{6} - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \right]$$

$$= 2 \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right]$$

$$= \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

(b) $y = 2 \tan^{-1} \left(\frac{x}{2} \right); 2 \tan \frac{y}{2} = x$



$$\text{Area} = \text{Area rect} - \int_0^{\pi/2} x dy$$

$$= 2 \cdot \frac{\pi}{2} - \int_0^{\pi/2} 2 \tan \frac{y}{2} dy$$

$$= \pi + 4 \left[\ln \left| \cos \frac{y}{2} \right| \right]_0^{\pi/2}$$

$$= \pi + 4 \left[\ln \frac{1}{2} - \ln 1 \right]$$

$$= (\pi - 2 \ln 2) \text{ or } 2$$

$$f(0.851325) = 0.001 \therefore \text{root between } 0.85 \text{ and } 0.851$$

∴ answer

13 a) 6

$$(1) y = x + \frac{9}{x}$$

$$y' = 1 - \frac{9}{x^2}$$

$$y' = 0$$

$$\therefore x = \pm 3$$

$$(3, 6), (-3, -6)$$

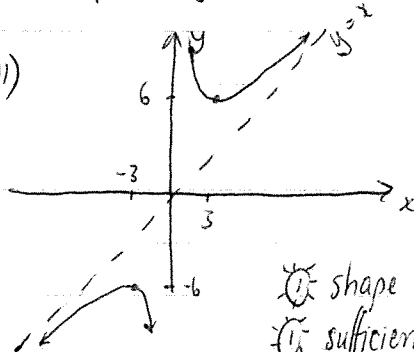
$$y'' = \frac{18}{x^3}$$

$$x = 3, y'' > 0 \cup \text{min}$$

$$x = -3, y'' < 0 \cap \text{max}$$

(ii) vertical: $x = 0$ oblique: $y = x$

(iii)



shape
sufficient
labelling

$$(b) \frac{t_{k+1}}{t_k} = \frac{{}^{18}C_{k+1} (7)^{17-k} (3)^{k+1}}{{}^{18}C_k (7)^{18-k} (3)^k}$$

$$= \frac{{}^{18}C_{k+1}}{{}^{18}C_k} \cdot \frac{3}{7}$$

$$= \frac{18-k}{k+1} \cdot \frac{3}{7}$$

$$\therefore \frac{54-3k}{7k+7} > 1$$

$$\therefore k < 4.7$$

$$\text{So } k = 4$$

$\therefore$ Choose the 5th

$$\text{Ans} = {}^{18}C_5 7^{13} 3^5$$

(c) 3

$$(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n \quad (1)$$

Differentiating: (2)

$$n(1+x)^{n-1} = C_1 + 2C_2 x + \dots + nC_n x^{n-1} \quad (2)$$

In (1) sub $x=1$

$$2^n = C_0 + C_1 + C_2 + \dots + C_n \quad (3)$$

In (2) sub $x=1$

$$n 2^{n-1} = C_1 + 2C_2 + \dots + nC_n \quad (4)$$

(1) (3) + (4)

$$n 2^{n-1} + 2 \cdot 2^{n-1} = C_1 + 2C_2 + 3C_3 + \dots + (n+1)C_n$$

$$\text{ie } (n+2) \cdot 2^{n-1} = C_1 + 2C_2 + 3C_3 + \dots + (n+1)C_n$$

as required.

(d) Try $n=1$: $1-1=0$ which is divisible by 6. TrueAssume true for $n=k$ ie $k^3 - k = 6M$, ($M \in \mathbb{Z}$)Prove true for $n=k+1$ ie $(k+1)^3 - (k+1) = 6N$; show $N \in \mathbb{Z}$

$$\text{LHS} = (k+1)^3 - (k+1)$$

$$= k^3 + 3k^2 + 3k + 1 - k - 1$$

$$= k^3 - k + 3k^2 + 3k$$

$$= 6M + 3(k^2 + k)$$

$$= 6 \left[M + \frac{k^2 + k}{2} \right]$$

$$= 6N, \text{ provided } k^2 + k \text{ is even.}$$

$$= \text{RHS}$$

$\therefore$ True for $n=k+1$

Since true for $n=1$ it will be true for $n=2$, and $n=3$ and so on.Hence true for all integers $n \geq 1$

structure of proof



If k is odd,
 k^2 is odd
the sum of 2 odds
is even

If k is even
 k^2 is even
the sum of 2 evens
is even

$\therefore k^2 + k$ is even
for all k .

14 a) 

$$(i) \frac{1}{2} \left[\tan^{-1} \frac{x}{2} \right]_{-2\sqrt{3}}^2 \quad \text{⊗}$$

$$= \frac{1}{2} \left[\tan^{-1} \frac{2}{2} - \tan^{-1} \left(\frac{-2\sqrt{3}}{2} \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} + \frac{\pi}{3} \right]$$

$$= \frac{7\pi}{24} \quad \text{⊗}$$

$$(ii) \text{ To prove } \tan^{-1} a - \tan^{-1} b = \tan^{-1} \left(\frac{a-b}{1+ab} \right)$$

first take tangents of both sides

$$\text{So prove } \tan[\tan^{-1} a - \tan^{-1} b] = \frac{a-b}{1+ab} \quad \text{⊗}$$

$$\text{LHS} = \frac{\tan(\tan^{-1} a) - \tan(\tan^{-1} b)}{1 + \tan(\tan^{-1} a) \tan(\tan^{-1} b)} \quad \text{⊗}$$

$$= \frac{a-b}{1+ab}$$

= RHS

$$b) \quad \text{⊗} \quad t=0, v=0, x=a$$

$$(i) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -n^2 x$$

$$\frac{1}{2} v^2 = \frac{-n^2 x^2}{2} + C \quad \text{⊗}$$

$$\text{sub } x=a, v=0 \Rightarrow C = \frac{n^2 a^2}{2}$$

$$\therefore \frac{1}{2} v^2 = \frac{-n^2 x^2}{2} + \frac{n^2 a^2}{2} \quad \text{⊗}$$

$$v^2 = -n^2 x^2 + n^2 a^2$$

$$v^2 = n^2 (a^2 - x^2)$$

$$(ii) v^2 = n^2 (a^2 - x^2)$$

$$\therefore v = n \sqrt{a^2 - x^2} \quad (v \text{ might be } + \text{ or } -)$$

$$\frac{dx}{dt} = -n \sqrt{a^2 - x^2}$$

$$\frac{dt}{dx} = \frac{1}{-n \sqrt{a^2 - x^2}} \quad \text{⊗}$$

$$t = \frac{1}{n} \int \frac{-1}{\sqrt{a^2 - x^2}} dx$$

$$= \frac{1}{n} \cos^{-1} \frac{x}{a} + C$$

$$\text{sub } t=0, x=0 \Rightarrow C=0$$

$$t = \frac{1}{n} \cos^{-1} \frac{x}{a}$$

$$\cos nt = \frac{x}{a} \therefore x = a \cos nt \quad \text{⊗}$$

-b) continued

$$(iii) v = -na \sin nt \quad \text{⊗}$$

$$(iv) \text{ greatest speed : } na \quad \text{⊗}$$

$$(v) \text{ greatest accel : } n^2 a \quad \text{⊗}$$

c)  (i) various methods

eg SSS or RHS or SAS

potential ingredients:

OP=OQ equal radii

AP=AQ equal tangents from external point

OA common

Angles at P, Q being right $\angle$ s

⊗ making headway

⊗ complete proof.

$$(ii) \angle PAO = \angle QAO$$

⊗ (Corresponding $\angle$ s in congruent Δ s)

⊗ So BX, XC subtend equal angles at circumference

$$\therefore BX = XC$$

$$\therefore X \text{ is midpoint of } BC$$